

$$\mathcal{L} = \mathcal{L}_{\text{ERP}} + 0.5 \mathcal{L}_{\text{slope}} + \lambda_p \mathcal{L}_{\text{peak}} + w \mathcal{L}_{\text{design}} + 0.1 \mathcal{L}_{\text{pad}}$$

$$\mathcal{L}_{\text{ERP}} = \text{mean} (\hat{y} - y)^2, \quad \mathcal{L}_{\text{slope}} = \text{mean} (\Delta \hat{y} - \Delta y)^2$$

$$\mathcal{L}_{\text{peak}} = \sum_{\text{peaks}} (\hat{y} - y)^2, \quad \lambda_p = 0.05 \text{ if } e > 0.8 E, \text{ else } 0$$

$$\mathcal{L}_{\text{design}} = \text{Huber}(\hat{a}, a), \quad \mathcal{L}_{\text{pad}} = \text{Huber}(\hat{z}, 0), \quad w = \min(1, (e + 1)/10)$$

Δ : difference between neighbouring frequencies; peaks: resonance peaks of the true ERP

a : resonator design (m, f_t, x, y) of each resonator; $\hat{a}$: design recovered from the true ERP by the inverse

z : extra padded values of the design (zero going forward); $\hat{z}$: their recovered values

y : true ERP; $\hat{y}$: predicted ERP; e : current epoch; E : total number of epochs